



Neat Enough to Be Green

A Randomised Vignette Experiment on Trailer-Load Classification at the Gate

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The problem

Every trailer arriving at a waste transfer station is classified at the gate, in a few seconds, as green waste or general waste. That call sets the fee and enters the council's landfill-diversion return. What the call is actually made on is nowhere recorded.

Research questions

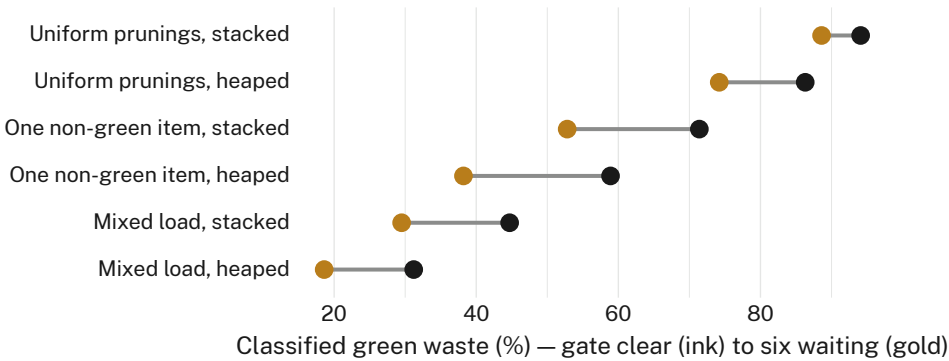
- Does composition decide the gate classification, or presentation?
- How far does queue pressure at the gate move the call?
- Is the classification stable enough to carry a diversion statistic?

Study design

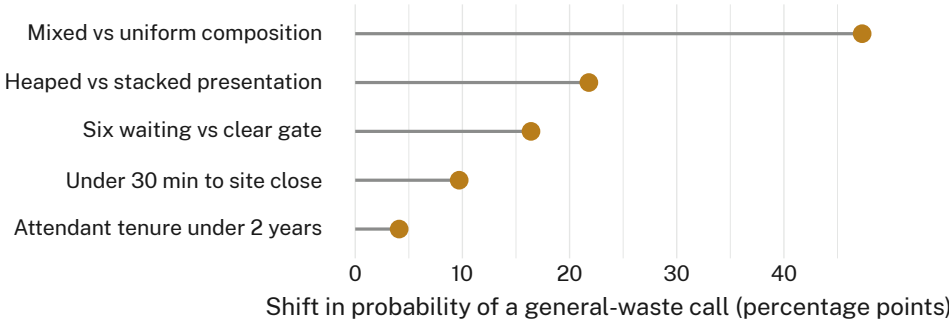
- 148 gate attendants · 12 randomised load vignettes each · 3 × 2 × 2 factorial · 6 councils · 9 weeks (1,776 judgements)
- Mixed-effects logistic model; attendant and site as random intercepts, with attendant tenure and minutes to site close recorded as covariates
- Vignette deck, factor levels, and the primary contrast pre-registered before recruitment; none adjusted post hoc

What we found

Presentation and queue state move the classification further than composition does. A stacked load carrying one non-green item is called green waste in 71.4% of judgements at a clear gate, and in 38.2% when the same composition arrives heaped with six vehicles waiting.



Green-waste classification rate by load composition and presentation, at a clear gate (ink) and with six vehicles waiting (gold). Across 1,776 judgements collected between March and May 2026, the queue effect is widest where composition is ambiguous: a stacked load carrying one non-green item falls 18.6 percentage points, against 5.5 for uniform prunings.



Estimated marginal effects from the fitted model. Composition remains the largest single term, but presentation and queue state together shift the call by 38.2 percentage points — more than four fifths of composition's own effect — and proximity to closing time adds a further 9.7 on its own.

Implications & takeaways

Results suggest the gate classification indexes how a load is presented, and how busy the gate is, at least as strongly as what the load contains. Where a diversion return aggregates gate calls unweighted, the resulting figure carries a throughput signal it is not read as carrying. Two of the four terms moving the call describe the gate rather than the trailer, and neither is recorded against the classification it produced. The instrument is stable within an attendant and unstable across queue state. Next: a paired gate-call and tipping-floor audit across a full reporting year.

Selected references

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We thank the gate staff of the participating transfer stations. Fieldwork was supported by a School of Continuous Improvement seed grant.

De-identified vignette responses only; no attendant's own gate log was accessed, and no council's diversion return was recalculated.

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